Avi Wong

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EDUCATION

University of California, Berkeley, Berkeley, CA — Expected December 2026
Master of Development Engineering — GPA: 4.0/4.0

Carnegie Mellon University, Pittsburgh, PA — May 2025
Bachelor of Science in Materials Science & Engineering — GPA: 4.0/4.0
Additional Major in Engineering & Public Policy

Relevant Courses: Challenges and Innovation in Labor Policy (auditing), Energy Regulation and Public Policy

PROFESSIONAL EXPERIENCE

Summer Intern, Local Clean Energy Alliance — June - August 2026

- Assisted in development of survey questions asking community members about cost and appliance preferences for low-income building electrification programs
- Combined geospatial data into a vulnerability index using ArcGIS; created interactive map to help staff identify the top 50 most vulnerable census tracts for survey outreach
- Delivered three-minute public comments advocating for policy and budgeting recommendations for a community choice aggregator

PROJECT EXPERIENCE

Environmental Life Cycle Assessment, UC Berkeley — September 2026 - present

- Retrieved greenhouse gas emissions and water consumption data from Federal LCA Commons flows and calculated cradle-to-gate emissions for natural gas electricity generation in Central California
- Calculated the energy consumption of data centers with different cooling systems across three cities with differing climates, accounting for transmission grid losses

Data Science for Energy, UC Berkeley — January - May 2026

- Modeled optimal economic dispatch of a gas generator and solar panels for a feeder network in a simulated power grid in the face of uncertain solar power generation using Python
- Modified previous model to include energy storage to help team compare estimated capital costs of lithium ion batteries and carbon cement supercapacitors

84-469 - Decision Science in International Relations — October - December 2024
Term Project - Health Communications During Prescribed Burn Smoke Events

- Applied prospect theory to predict how rural residents may respond to increased smoke depending on the framing of communications received
- Designed multiple communication strategies based on trust in government sources; advocated for residents to help design communications and local smoke plans

Undergraduate Research, Carnegie Mellon University — June - August 2024
Validating Molecular Models for Simulating Step-Growth Polymerization of Linear Polymers

- Composed a 3-page grant proposal; praised for communicating technical concepts well to laypeople
- Developed presentable, accessible line graphs and histograms from raw simulation data
- Conducted statistical tests using SciPy to verify that data's weight fraction distribution matched theoretical distribution
- Presented poster to university's undergraduate research symposium

19-351 - Applied Methods in Technology-Policy Analysis, Carnegie Mellon — January - May 2024
Strategic Framework for Broadband Expansion in Pennsylvania

- Developed nonlinear model to estimate per-site cost of connecting rural schools and libraries in Pennsylvania to fiber-based broadband
- Conducted sensitivity analysis in Excel to identify the factors that cause the most variation in the costs of different broadband technologies

SKILLS

Software: Python, CVXPY, ArcGIS, Excel/Google Sheets, Canva, Google Slides
Languages: English (native), Spanish (proficient), Cantonese (beginner)